

Week 1 Objective: Identify literature and develop Outline/Structure

Activity	Status	Literature	Who?
Review literature on class imbalance and techniques used to improve model performance	Class Imbalance - Nature and extent of class imbalance in fraud detection datasets - Impact of class imbalance on model performance - Effects on accuracy, precision, recall, and F1-score - Challenges in minority class prediction Techniques Used to Improve Model Performance - Random oversampling - Random undersampling - Synthetic Minority Oversampling Technique (SMOTE) - Adaptive Synthetic Sampling (ADASYN) - Cost-sensitive learning - Class weighting - Ensemble learning methods - Threshold optimization - Feature engineering and feature selection - Hyperparameter tuning - Cross-validation techniques - Data augmentation methods	Adebayo et al., 2026; Albalawi et al., 2025; Al-dahasi et al., 2025; Almazroi and Ayub, 2023; Boulrieris et al., 2024; Cai and Yang, 2026; Chen, 2025; Compagnino et al., 2025; Luong and Xie, 2026.	Edem/Omotayo
Examination of the effects of concept drift and evolving fraud patterns on the effectiveness of machine learning fraud detection	- Concept Drift, Evolving Fraud Patterns, and Adversarial Evolution - Concept drift in fraud detection - Types of concept drift (sudden, gradual, incremental, recurring) - Evolution of fraud patterns and tactics - Behavioural changes in fraudsters - Adversarial evolution and fraud evasion techniques - Impact of concept drift on model accuracy and reliability - Model degradation over time	Adebayo et al., 2026; Cai and Yang, 2026; Compagnino et al., 2025; Gupta and Babu, 2025; Luong and Xie, 2026	Edem/Omotayo

models. (And Adversarial Evolution)	• Continuous learning and model retraining • Incremental and online learning approaches • Adaptive and dynamic machine learning models • Ensemble updating techniques • Transfer learning for evolving fraud environments • Continuous model monitoring and performance evaluation • Best practices for maintaining long-term model effectiveness.		
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